

Outputs will be saved to: /content/outputs
Using device: cuda

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SPHERE EXPERIMENT (d=8, D=64, N_train=100,000)
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[1] Generating sphere data...
    Train: torch.Size([100000, 64]), Test:
torch.Size([500, 64])
    Data range: [-0.547, 0.560]
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[2] Training Metric Matching MLP...
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    Model parameters: 950,656
Epoch   50/1000 | Loss: -967.9027
Epoch  100/1000 | Loss: -967.2594
Epoch  150/1000 | Loss: -967.0355
Epoch  200/1000 | Loss: -965.1841
Epoch  250/1000 | Loss: -966.6845
Epoch  300/1000 | Loss: -968.8160
Epoch  350/1000 | Loss: -968.1743
Epoch  400/1000 | Loss: -968.1752
Epoch  450/1000 | Loss: -968.8217
Epoch  500/1000 | Loss: -970.0234
Epoch  550/1000 | Loss: -966.1316
Epoch  600/1000 | Loss: -967.7572
Epoch  650/1000 | Loss: -969.3723
Epoch  700/1000 | Loss: -967.4133
Epoch  750/1000 | Loss: -965.9098
Epoch  800/1000 | Loss: -969.5879
Epoch  850/1000 | Loss: -967.6802
Epoch  900/1000 | Loss: -969.3307
Epoch  950/1000 | Loss: -966.9185
Epoch 1000/1000 | Loss: -967.2692
    Final training loss: -967.2692
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[3] Evaluating tangent space prediction...
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    Metric Matching: Frobenius dist = 3.7526 (best
 $\epsilon=0.25$ )
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[4] Evaluating k-NN CDC baseline...
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    k-NN CDC: Frobenius dist = 3.7827 (k=128,  $\epsilon=0.05$ )
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[5] Computing eigenvalue spectra...

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RESULTS SUMMARY

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Metric Matching Frobenius dist: 3.7526
k-NN CDC Frobenius dist: 3.7827
Improvement: +0.8%
(Paper reports ~46% improvement for smaller datasets)

Generating figures...

Figure saved to: /content/outputs/
metric_matching_results.png

Saved figure at: /content/outputs/
metric_matching_results.png

[Demo] Computing intrinsic gradient of a test function...

Point norm: 1.0000
Intrinsic gradient norm: 0.1675
(Intrinsic grad is in tangent space, so $\langle \text{grad}, \text{point} \rangle \approx 0$)
 $\langle \text{intrinsic_grad}, \text{point} \rangle = 0.158646$ (should be ~ 0 on sphere)

✓ Reproduction complete. Results saved to outputs/
metric_matching_results.png

Reproduction: Riemannian Metric Synthetic Sphere

